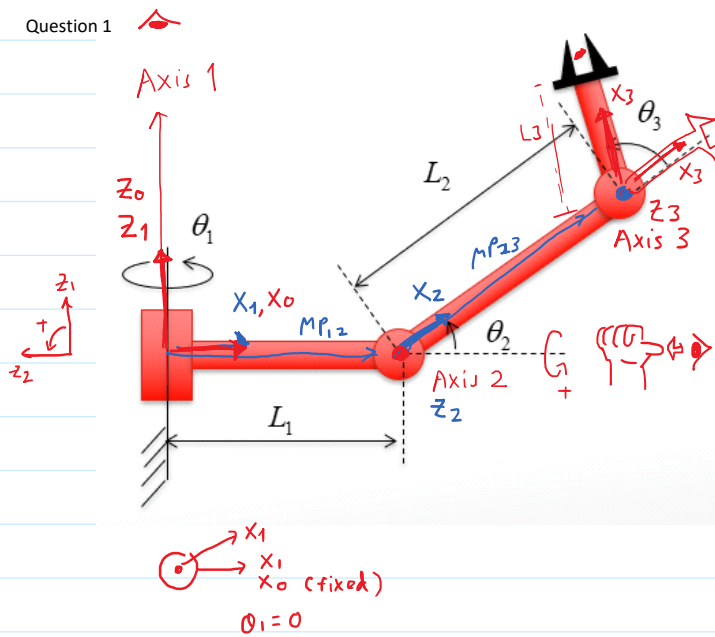


Question 1



1) Step 1: Draw axes

2) Step 2: Mutual perpendicular

3) Step 3: Frames $\{1\}$ to $\{n-1\}$

4) Step 4: Frame $\{0\}$

5) Step 5: Frame $\{n\} \rightarrow \{3\}$

(X_1 aligned with X_2 when $\theta_3 = 0$)

6) Step 6: Link lengths a_{i-1}

$a_0 =$ distance from Z_0 to Z_1 along $X_0 = 0$

$a_1 =$ " Z_1 Z_2 " $X_1 = L_1$

$a_2 =$ " Z_2 Z_3 " $X_2 = L_2$

7) Step 7: Link twists α_{i-1}

$\alpha_0 = \angle (Z_0, Z_1)$ about $X_0 = 0$

$\alpha_1 = \angle (Z_1, Z_2)$ about $X_1 = 90^\circ$

$\alpha_2 = \angle (Z_2, Z_3)$ about $X_2 = 0$

8) Step 8: Link offsets d_i

$d_1 =$ distance from X_0 to X_1 along $Z_1 = 0$

$d_2 =$ " X_1 to X_2 " $Z_2 = 0$

$d_3 =$ distance " X_2 to X_3 " $Z_3 = 0$

9) Step 9: Joint angles θ_i

$\theta_1 = \angle (X_0, X_1)$ about $Z_1 =$ variable

$\theta_2 = \angle (X_1, X_2)$ about $Z_2 =$ variable

$\theta_3 = \angle (X_2, X_3)$ about $Z_3 =$ variable

10) DH Table:

i	α_{i-1}	a_{i-1}	d_i	θ_i	
1	0	0	0	θ_1	$\leftarrow$
2	90	L_1	0	θ_2	$\leftarrow$
3	0	L_2	0	θ_3	$\leftarrow$

11) Transformation matrices:

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ ca_{i-1}s\theta_i & ca_{i-1}c\theta_i & -sa_{i-1} & -sa_{i-1}d_i \\ sa_{i-1}s\theta_i & sa_{i-1}c\theta_i & ca_{i-1} & ca_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

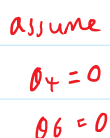
$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & L_1 \\ 0 & 0 & -1 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & L_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where $C_i = \cos(\theta_i)$, $S_i = \sin(\theta_i)$
 $C_{23} = \cos(\theta_2 + \theta_3)$, $S_{23} = \sin(\theta_2 + \theta_3)$

Step 13) ${}^0P = {}^0T \cdot {}^3P = \begin{bmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{bmatrix} \begin{bmatrix} L_1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Question 2


$$\alpha_5 = \chi(z_5, z_6) \text{ about } X_5 = -\phi$$

④ 4 }

Step 8) Link offsets :

$d_4 =$ distance from X_3 to X_4 along $Z_4 = 0$
 $d_5 =$ " X_4 to X_5 " $Z_5 = 0$
 $d_6 =$ " X_5 to X_6 " $Z_6 = 0$

Step 9) Joint angles

θ_4
 θ_5
 θ_6 } variable

Step 10) DH Table

i	α_{i-1}	a_{i-1}	d_i	θ_i	
4	0	0	0	θ_4	$\leftarrow$
5	ϕ	0	0	θ_5	$\leftarrow$
6	$-\phi$	0	0	θ_6	

Step 11) Transformation matrices

$${}^{i-1}_i T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4 T = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & 0 \\ s\theta_4 & c\theta_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_5 T = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & 0 \\ c\phi s\theta_5 & c\phi c\theta_5 & -s\phi & 0 \\ s\phi s\theta_5 & s\phi c\theta_5 & c\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6 T = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ c\phi s\theta_6 & c\phi c\theta_6 & s\phi & 0 \\ -s\phi s\theta_6 & -s\phi c\theta_6 & c\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$s(-\phi) = -s\phi$$

$$c(-\phi) = c\phi$$

Finally: ${}^3_6 T = {}^3_4 T \cdot {}^4_5 T \cdot {}^5_6 T$
 = try it yourself.

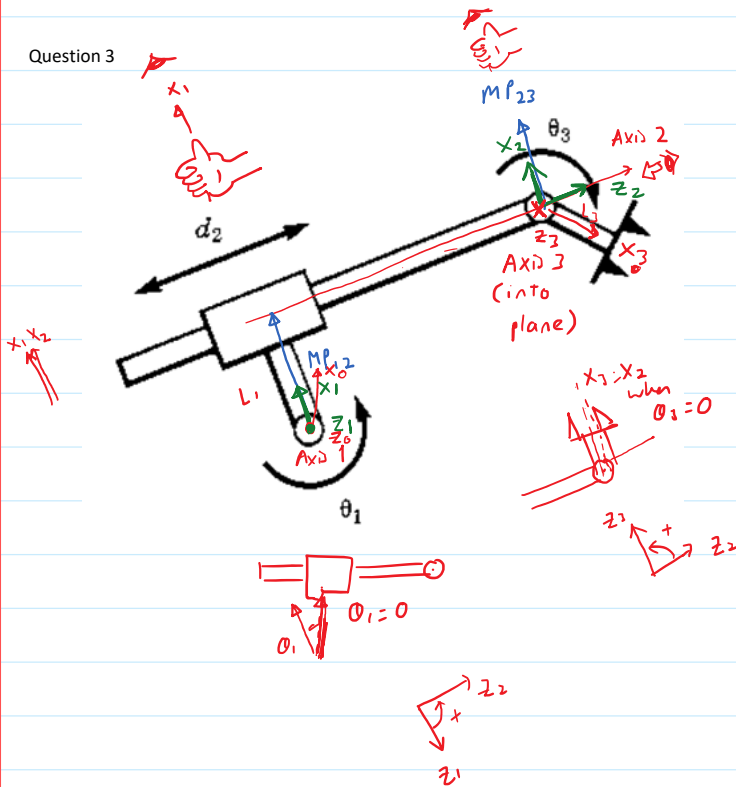
Check : if $\theta_1=0, \theta_2=0, \theta_3=0$ $\Rightarrow$
 $\theta_4=0, \theta_5=0, \theta_6=0$
 (errata in video)

$${}^3_6 T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 12) ${}^6_P = \begin{bmatrix} 0 \\ 0 \\ L \end{bmatrix}$

Step 13) ${}^3_P = \underline{\underline{{}^3_6 T \cdot {}^6_P}}$

Question 3



1) Draw axes

2) Mutual perpendiculars

3) $\{1\}$ to $\{n-1\} = \{2\}$

4) $\{0\}$

5) $\{n\} = \{3\}$

$X_3 = X_2$ when $\theta_3 = 0$

6) a_{i-1}

$a_0 =$ distance from Z_0 to Z_1 along $X_0 = 0$

$a_1 =$ " Z_1 to Z_2 " $X_1 = L_1$

$a_2 =$ " Z_2 to Z_3 " $X_2 = 0$

7) α_{i-1}

$\alpha_0 = \angle (Z_0, Z_1)$ about $X_0 = 0$

$\alpha_1 = \angle (Z_1, Z_2)$ " $X_1 = 90^\circ$

$\alpha_2 = \angle (Z_2, Z_3)$ " $X_2 = 90^\circ$

8) d_i

$d_1 =$ distance from X_0 to X_1 along $Z_1 = 0$

$d_2 =$ " X_1 to X_2 " $Z_2 = \text{Variable}$

$d_3 =$ " X_2 to X_3 " $Z_3 = 0$

9) θ_i

$\theta_1 = \text{Variable}$

$\theta_2 = \angle (X_1, X_2)$ about $Z_2 = 0$

$\theta_3 = \text{Variable}$

10) DH Table

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	90	L_1	d_2	0
3	90	0	0	θ_3

11) Transform matrices

$${}^{i-1}_i T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_1 T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2 T = \begin{bmatrix} 1 & 0 & 0 & L_1 \\ 0 & 0 & -1 & -d_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3 T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

errata in video

$${}^0_3T = {}^0_1T \cdot {}^1_2T \cdot {}^2_3T$$

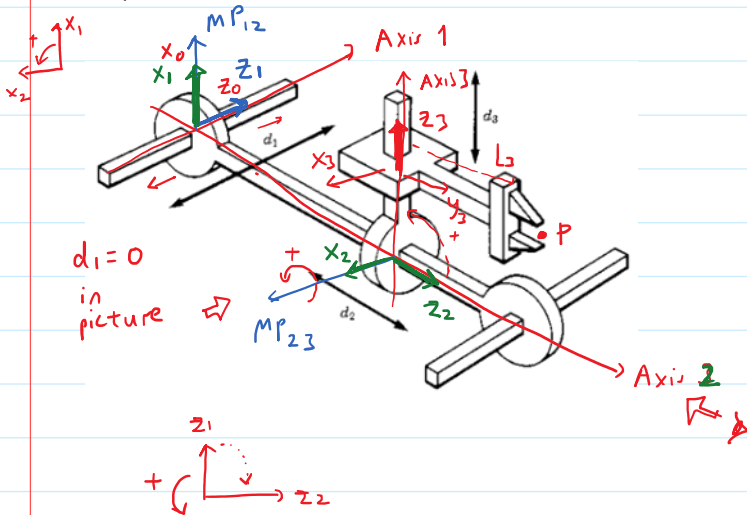
$$= \begin{bmatrix} c_1 c_3 + s_1 s_3 & -c_1 s_3 + s_1 c_3 & 0 & L_1 c_1 + d_2 s_1 \\ s_1 c_3 - c_1 s_3 & -s_1 s_3 - c_1 c_3 & 0 & L_1 s_1 - d_2 c_1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$12) {}^3P = [L_3 \ 0 \ 0]^T$$

$$13) {}^0P = \underline{\underline{{}^0_3T \cdot {}^3P}}$$



Question 4



1) Axes

2) Mutual h

3) $\{1\}$ to $\{2\}$

4) $\{0\}$

5) $\{3\}$ X_3 aligned with X_2 when $d_3 = 0$

6) a_{i-1}

$a_0 =$ distance from Z_0 to Z_1 along $X_0 = 0$

$a_1 =$ " Z_1 to Z_2 " $X_1 = 0$

$a_2 =$ " Z_2 to Z_3 " $X_2 = 0$

7) α_{i-1}

$\alpha_0 = \angle (Z_0, Z_1)$ about $X_0 = 0$

$\alpha_1 = \angle (Z_1, Z_2)$ about $X_1 = -90^\circ$

$\alpha_2 = \angle (Z_2, Z_3)$ about $X_2 = 90^\circ$

8) d_i

$d_1 =$ distance from X_0 to X_1 along $Z_1 =$ variable

d_2 } variable, prismatic

d_3

9) θ_i

$\theta_1 = \angle (X_0, X_1)$ about $Z_1 = 0$

$\theta_2 = \angle (X_1, X_2)$ about $Z_2 = 90^\circ$

$\theta_3 = \angle (X_2, X_3)$ about $Z_3 = 0$

10) DH Table

i	α_{i-1}	a_{i-1}	d_i	θ_i	
1	0	0	d_1	0	↖
2	-90°	0	d_2	90°	↖
3	90°	0	d_3	0	

11) Transformation matrices

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_1T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -d_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_3T = {}^0_1T \cdot {}^1_2T \cdot {}^2_3T = \begin{bmatrix} 0 & 0 & 1 & d_3 \\ 0 & 1 & 0 & d_2 \\ -1 & 0 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$12) {}^3P = \begin{bmatrix} 0 \\ L_3 \\ 0 \end{bmatrix}$$

$$13) {}^0P = \underline{\underline{{}^0_3T \cdot {}^3P}}$$